

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 10: Circles

Every formula and result of the chapter on one sheet

1	Tangent and radius	Tangent $\perp$ radius at the point of contact
2	Tangents from an external point	$PA = PB$
3	Length of a tangent	$PT = \sqrt{d^2 - r^2}$, where d is the distance of the point from the centre
4	Distance of the point from the centre	$d = \sqrt{PT^2 + r^2}$
5	Angle between two tangents	$\angle APB + \angle AOB = 180^\circ$
6	Line from the point to the centre	OP bisects $\angle APB$ and $\angle AOB$
7	Number of tangents from a point	Inside: 0; on the circle: 1; outside: 2
8	Quadrilateral circumscribing a circle	$AB + CD = AD + BC$
9	Chord touching the inner of two concentric circles	Length $= 2\sqrt{R^2 - r^2}$
10	Triangle with an incircle	$AF = AE = s - a$, $BD = BF = s - b$, $CD = CE = s - c$, where s is the semi-perimeter
11	Radius of the incircle of a right-angled triangle	$r = \frac{a + b - c}{2}$, where c is the hypotenuse